A Variational Bayes Mixture Model for Unsupervised Image Segmentation

Abstract

A Variational Bayes mixture model for unsupervised image segmentation is described in this paper. The reversible jump Markov chain Monte Carlo (RJMCMC) provides a principled way to determine the model structure and the model parameters jointly. RJMCMC methods can, in principle, find exact results, but they require a huge amount of time for computation. Recently, an alternative approach called the Variational Bayes (VB) method has been proposed. This method defines the free energy for a trial probability distribution, which approximates a joint posterior probability distribute over model parameters and hidden variables. The maximum of the free energy gives the log evidence for an observed data set. We introduce the VB method into modeling the Gaussian mixture for image segmentation. The task of identifying the mixture component parameters, labeling the underlying class at each site, and determining the number of components is accomplished using the batch VB scheme, in which identifying the parameters and labeling the class can be done analytically in an iterative EM-like algorithm, and determining the number of components is in sequential procedure with split and merge operations. We develop new split and merge methods. The methods can extend one of Richardson and Green from one-dimensional to high-dimensional data space. We report some results on color image segmentation in this paper.

1. Introduction

Image segmentation has always played a central role in computer vision. Unsupervised segmentation may be defined as the task of dividing an image into some regions based on some measure of similarity of pixels. It makes the pixels in each region have uniform similarity as much as possible. Each distinct model from which pixels are realized is referred to as a class. An efficient approach to measuring the similarity of pixels is probabilistic model. There are several unsupervised image segmentation methods based on probabilistic models. These methods typically use Gaussian mixtures to model the classes and the Pott's model on the class labels to encourage spatial connectivity. In this paper, we restrict our image model to the Gaussian mixture model.

Gaussian mixtures are ubiquitous probabilistic models for density estimation due to they benefiting from the analytical tractability and asymptotic properties of the Gaussian distribution, and being a universal approximator for continuous densities. The associated model parameters are then estimated in an approximate MAP setting or using maximum likelihood (ML) estimation, with expectation maximization (EM)[5] algorithm or more recent varieties. The fundamental problems of the ML method are overfitting and the inability to determine the model structure.

In the most general case, not only the parameters of each class, but the number of classes, are known a prior. So, we need to refer to some statistical procedure to infer the number of classes. This may be treated as a model section problem. The Bayesian framework is proposed to deal with model section [3]. The evidence gives a criterion for the model selection. However, computations in the Bayesian framework can seldom be performed exactly due to the need to integrate over parameters. Some approximations therefore must be made, the main schemes being Markov Chain Monte Carlo (MCMC) methods [1] and the Laplace approximation method [9]. The former can, in principle, find exact results but typically requires vast computational costs. The latter has lower complexity but is only valid for a large sample limit.

MCMC methods play a central role in modern Bayesian computation. To apply them for varying-dimension problem have been discussed [7,8]. One approach, termed reversible jump MCMC (RJMCMC), is elaborated in Green [7]. In brief, RJMCMC is a random-sweep Metropolis-Hastings method including the split-merge operations and the birth-death process. The method is attractive since it can preferably deal with problems about parameters estimation and model selection

together. Richardson and Green [9] applied RJMCMC for performing a fully Bayesian analysis of data from a finite mixture distribution with an unknown number of components. But, in one hand, it is computationally costly, in the other hand, it requires data with one-dimension, and suppose that the mean vector satisfies certain order relation under estimating the parameters. In order to overcome these problems, Stephens [12] uses the birth-dearth process instead of the reversible-jump methodology, and describes an alternative MCMC, which is called the birth-dearth MCMC (BDMCMC). But they still take a huge amount time for computation. In order to reduce computation, the data-driven techniques are introduces into the MCMC methods with applications in image segmentation and object recognition [4,13,15]. This makes tradeoff between computational cost and power. These methods incorporate the data-driven techniques into the split-merge operations within the reversible jump.

Recently, Variational Bayes (VB) approach has been proposed [16,2,6,11]. This approach defines the free energy for a trial probability distribution, which approximates a joint posterior probability distribution over model parameters and hidden variables. The maximum of the free energy gives the log evidence for an observed data set. The posterior distributions are obtained via an iterative EM-like algorithm whose convergence is guaranteed.

Our work is inspired by the VB method. In this paper, we introduce the VB method into the Gaussian mixture model for unsupervised image segmentation. Sequential model selection procedure has been adopted by combing the VB method and the split and merge algorithm. Our split and merge algorithm differs from ones by Ueda et al. [12] or Zhu et al. [13], and extends that of Richardson and Green [8] to high-dimensional data space.

2. Problem Formulation

2.1. Basis Definitions

Let $\Xi = \{i = (a,b) : 0 \le a \le A, 0 \le b \le B\}$ be the site lattice for an image of dimension $A \times B$ and class parameter $M \in \{1,2,\cdots,P\}$ denote the (unknown) number of classes. At each pixel i in the image, we compute a *D*-dimensional feature vector $\boldsymbol{x}_i$, which encapsulates position, colour and texture information, and use $y_i \in \{1,2,\cdots,M\}$ to indicate from class of the pixel i came. We write $\boldsymbol{X} = \{\boldsymbol{x}_i : i \in \Xi\}$ for the observed image data, and $\boldsymbol{Y} = \{y_i : i \in \Xi\}$ for a realization of a *M*-class label field over Ξ.

2.2. Image Model

Our image model is that each pixel is produced by a Gaussian density. These Gaussian densities, each of which is called as component, constitute a Gaussian mixture model. Consider a mixture model with $M > 1$ components in R^D for $D \ge 1$:

$$p(\boldsymbol{x}) = \sum_{m=1}^{M} p(m)\, p_m(\boldsymbol{x}|m) = \sum_{m=1}^{M} p_m\, p_m(\boldsymbol{x}|m) \qquad \forall \boldsymbol{x} \in R^D, \quad (1)$$

where $p_m \in (0,1)$ ($\forall m = 1,2,\cdots,M$) are the mixing proportions subject to $\sum_{m=1}^{M} p_m = 1$. Gaussian mixture model means that in mixture model, each component density of which $p_m(\boldsymbol{x}|m)$ is a normal probability distribution:

$$p_m(\boldsymbol{x}|m) = \frac{1}{(2\pi)^{D/2} \det(\Sigma_m)^{1/2}} \exp\left\{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_m)^{\mathrm{T}} \boldsymbol{S}_m^{-1} (x-\boldsymbol{\mu}_m)\right\}$$

We encapsulate these parameters into a parameter vector, writing the parameters of each component as $q_l = (\boldsymbol{\mu}_l, \boldsymbol{S}_l)$, to get $\Theta = (p_1, p_2, \cdots, p_M, q_1, q_2, \cdots, q_M)$. Then, (1) could be rewritten as

$$p(\boldsymbol{x}|\Theta) = \sum_{m=1}^{M} p_m\, p_m(\boldsymbol{x}|q_m). \quad (2)$$

If we knew the component from which $\boldsymbol{x}$ came, then it would be simple to determine the parameters Θ. Similarly, if we knew the parameters Θ, we could determine the component that is most likely to have produced $\boldsymbol{x}$. The difficulty is that we know neither. However, the EM algorithm could be introduced to deal with the difficulty by utilizing the missing data strategy.

In our image model, the set $\boldsymbol{Z} = (\boldsymbol{X}, \boldsymbol{Y})$ consists of observed data $\boldsymbol{X}$ and missing (hidden) data $\boldsymbol{Y}$. $\boldsymbol{Z}$ and $\boldsymbol{X}$ are called complete data and incomplete data, respectively. Assume that the joint probability density of $\boldsymbol{Z}$ is parametrically given as $p(\boldsymbol{X}, \boldsymbol{Y}; \Theta)$. The maximum likelihood estimate of Θ is a value of Θ that maximizes the incomplete data log-likelihood function:

$$L(\Theta;\boldsymbol{X}) = \log p(\boldsymbol{X}|\Theta) = \log \int p(\boldsymbol{X},\boldsymbol{Y}|\Theta) d\boldsymbol{Y}$$

The EM algorithm consists of E-step and M-step, which is to maximize the incomplete data log-likelihood function by iteratively maximizing the expectation of the complete data log-likelihood function $L_c(\boldsymbol{X};\Theta) = \log p(\boldsymbol{X},\boldsymbol{Y}|\Theta)$.

3. VB Learning

In the Bayesian method, a set of models, $\{M\}$, where M denotes a model with a fixed structure, is considered. A probability distribution for a model M with a model parameter Θ is denoted by $P(\boldsymbol{x}|\Theta,\mathsf{M})$, where $\boldsymbol{x}$ represents an observed stochastic variable. For a set of observed data, $\boldsymbol{X} = \{\boldsymbol{x}_i : i \in \Xi\}$, and a prior probability distribution $P(\Theta|\mathsf{M})$, the Bayesian method calculates the posterior probability over the parameter,

$$P(\Theta|\boldsymbol{X},\mathsf{M}) = \frac{P(\boldsymbol{X}|\Theta,\mathsf{M})P(\Theta|\mathsf{M})}{P(\boldsymbol{X}|\mathsf{M})}$$

Here, the data likelihood is defined by

$$P(\boldsymbol{X}|\Theta,\mathsf{M}) = \prod_{i\in\Xi} P(\boldsymbol{x}_i|\Theta,\mathsf{M})$$

and the evidence for a model M is defined by

$$P(\boldsymbol{X}|\mathsf{M}) = \int d\Theta P(\boldsymbol{X}|\Theta,\mathsf{M})P(\Theta|\mathsf{M}). \quad (3)$$

The posterior probability over model structure is given by

$$P(\mathsf{M}|\boldsymbol{X}) = \frac{P(\boldsymbol{X}|\mathsf{M})P(\mathsf{M})}{\sum_{\mathsf{M}'} P(\boldsymbol{X}|\mathsf{M}')P(\mathsf{M}')}.$$

3.1. Variational Bayes (VB) method

In the following, we will calculate the evidence for a given model by using the VB method and the explicit model structure dependence is omitted. The integration over model parameters in (3) penalizes complex models with more degrees of freedom. This integration, however, is often difficult to perform. In order to evaluate this integration, the VB method introduces a trial probability distribution $q(\Theta,\boldsymbol{Y})$, which approximates the posterior probability distribution over the model parameter Θ and the hidden variables $\boldsymbol{Y} = \{\boldsymbol{y}_i : i \in \Xi\}$:

$$P(\Theta,\boldsymbol{Y}|\boldsymbol{X}) = \frac{P(\boldsymbol{X},\boldsymbol{Y}|\Theta)P_0(\Theta)}{P(\boldsymbol{X})}$$

where $P_0(\Theta)$ denotes a prior distribution for the model parameters and the probability distribution for a complete data set $(\boldsymbol{X},\boldsymbol{Y})$ is given by

$$P(\boldsymbol{X},\boldsymbol{Y}|\Theta) = \prod_{i\in\Xi} P(\boldsymbol{x}_i,\boldsymbol{y}_i|\Theta).$$

The KL-divergence between the trial distribution $q(\Theta,\boldsymbol{Y})$ and the true posterior distribution $p(\Theta,\boldsymbol{Y}|\boldsymbol{X})$ is given by

$$KL(q\|p) = \int d\Theta d\boldsymbol{Y} q(\Theta,\boldsymbol{Y}) \times \log\left(\frac{q(\Theta,\boldsymbol{Y})}{p(\Theta,\boldsymbol{Y}|\boldsymbol{X})}\right)$$
$$= \log p(\boldsymbol{X}) - F(\boldsymbol{X},q), \quad (4)$$

where the free energy $F(\boldsymbol{X},q)$ is defined by

$$F(\boldsymbol{X},q) = \int d\Theta d\boldsymbol{Y} q(\Theta,\boldsymbol{Y}) \times \log\left(\frac{p(\boldsymbol{X},\boldsymbol{Y}|\Theta)P_0(\Theta)}{q(\Theta,\boldsymbol{Y})}\right). \quad (5)$$

Therefore, the true posterior distribution is obtained by maximizing the free energy w.r.t. the trial distribution $q(\Theta,\boldsymbol{Y})$. The maximum of the free energy is equal to the log evidence:

$$\log p(\boldsymbol{X}) = \max_q F(\boldsymbol{X},q) \geq F(\boldsymbol{X},q).$$

The above equation implies that the lower bound for the log evidence can be evaluated by using some trial posterior distribution $q(\Theta,\boldsymbol{Y})$.

In the VB method, trial posterior distributions are assumed to be factorized as

$$q(\Theta,\boldsymbol{Y}) = q(\Theta)q(\boldsymbol{Y}). \quad (6)$$

We can rewrite the function $\boldsymbol{X}$ as

$$F(\boldsymbol{X},q) = \left\langle \log \frac{p(\boldsymbol{X},\boldsymbol{Y}|\Theta)}{q(\boldsymbol{Y})} \right\rangle_{\boldsymbol{Y},\Theta} - KL[q(\Theta)\|p(\Theta)],$$

where the average in the first term on the r.h.s. is taken w.r.t. $q(\Theta,\boldsymbol{Y})$. The first term corresponds to the averaged likelihood. The second term is the KL distance between the prior and posterior over the parameters. We try to obtain a lower-bound estimate for the log evidence $\log(p(\boldsymbol{X}))$ by maximizing the free energy $F(\boldsymbol{X},q)$ w.r.t. factorized $q(\Theta,\boldsymbol{Y})$ in (6).

The free energy can be maximized by alternately maximizing the free energy w.r.t. $q(\Theta)$ and $q(\boldsymbol{Y})$. This process results in an iterative algorithm directly analogous to ordinary EM.

In the VB E-step, the free energy is maximized w.r.t. $q(\boldsymbol{Y})$ under the condition $\int d\boldsymbol{Y} q(\boldsymbol{Y}) = 1$, while $q(\Theta)$ is fixed. We compute the posterior over the hidden variables by solving $dF/dq(\boldsymbol{Y}) = 0$ to get

$$q(\boldsymbol{Y}) \propto e^{\langle \log p(\boldsymbol{X},\boldsymbol{Y}|\Theta)\rangle_\Theta},$$

where the average is taken w.r.t. $q(\Theta)$.

In the VB M-step, the free energy is maximized w.r.t. $q(\Theta)$ under the condition $\int d\Theta q(\Theta) = 1$, while $q(\boldsymbol{Y})$ obtained in the VB E-step is fixed. We compute the posterior over the parameters by solving $dF/dq(\Theta) = 0$ to get

$$q(\Theta) \propto e^{\langle \log p(\boldsymbol{X},\boldsymbol{Y}|\Theta)\rangle_Y} p_0(\Theta)$$

where the average is taken w.r.t. $q(\boldsymbol{Y})$.

3.2. Variational Bayes mixture models

We now applied the VB method to the Gaussian mixture model [2]. We use conjugate priors on the parameters $\Theta = \{(p_m, m_m, \Gamma_m) : m = 1, \cdots, M\}$, where Γ_m denotes the precision (inverse covariance Σ_m) matrix. We find that the parameters posterior for a fixed M factorizes into $q(\Theta) = q(\{p_m\})\prod_m q(\boldsymbol{\mu}_m, \Gamma_m)$. Assume that the mixing proportions are jointly Dirichlet, and the means and precision are Normal-Wishart. The posteriors are obtained by the following iterative EM-like algorithm.

E-step. The label posterior factorizes over the pixels $q(\boldsymbol{Y}|\boldsymbol{X}) = \prod_n q(y_n|\boldsymbol{x}_n)$. For pixel n, we first initialize $\bar{p}_s$, $\bar{\boldsymbol{\mu}}_s$ and $\bar{\Gamma}_s$ to appropriate values, then compute $q(y_n|\boldsymbol{x}_n)$ by

$$g_s^n = q(y_n = s|x_n) \propto \tilde{p}_s \tilde{\Gamma}_s \times e^{-(\boldsymbol{x}_n - \bar{\boldsymbol{\mu}}_s)^T \bar{\Gamma}_s (\boldsymbol{x}_n - \bar{\boldsymbol{\mu}}_s)/2} e^{-D/N\bar{p}_s},$$

where $s \in \{1,2,\cdots,M\}$, $N = |\Xi|$, $\tilde{p}_s = \exp(\Psi(N\bar{p}_s + 1) - \Psi(N + M))$, $\tilde{\Gamma}_s = (N\bar{p}_s/2)^{-D}|\bar{\Gamma}_s|\exp(\Psi(N\bar{p}_s/2))$, and Ψ is the psi (digamma) function.

M-step. We compute new $\bar{p}_s$, $\bar{\boldsymbol{\mu}}_s$ and $\bar{\Gamma}_s$ by

$$\bar{p}_s = \frac{1}{N}\sum_{n=1}^N g_s^n,$$

$$\bar{\boldsymbol{\mu}}_s = \frac{1}{N\bar{p}_s}\sum_{n=1}^N g_s^n \boldsymbol{x}_n,$$

$$\bar{\Sigma}_s = \frac{1}{N\bar{p}_s}\sum_{n=1}^N g_s^n (x_n - \bar{\boldsymbol{\mu}}_s)(x_n - \bar{\boldsymbol{\mu}}_s)^T.$$

Once the posterior distributions over the parameters and label conditioned on structure have been obtained, the posterior distribution over model structure $q(M)$ is given by

$$q(M|\boldsymbol{X}) \propto \exp\left(\left\langle \log \frac{p(\boldsymbol{X},\boldsymbol{Y},\Theta|M)}{q(\boldsymbol{Y},\Theta|M)}\right\rangle_{\boldsymbol{Y},\Theta|M}\right),$$

where the average is taken w.r.t. $q(\boldsymbol{Y},\Theta|M)$.

4. Sequential model section

In the usual Bayesian methods for model section, one develops a set of models with different structure and calculates the evidence for each model. Then, the model with the highest evidence is selected or the average over these models is taken. An alternative approach is to search better model structure by changing model structure sequentially. Sequential model selection processes have been proposed based on the VB algorithm [6,11] or the MCMC method [9,12]. In the VB formulation, we adopt sequential model selection method with the merge-and-split operations in this paper.

The idea of performing split-and-merge (SM) operation has been successfully applied to clustering and vector quantization.

Recently, Ueda et al. [14] applied the SM to the EM algorithm, and proposed the split-and-merge EM (SMEM) algorithm to improve the local maximum problem of the EM algorithm for the parameters estimation in a finite mixture model. In the SMEM algorithm, the SM operations are simultaneously performed so that the total number of mixture components is unchanged. Therefore, the SMEM algorithm is unable to determine the model structure. In this paper, we use the split and merge operations for model selection in the VB framework. We adopt the split and merge strategies in [11].

In the SMEM method, the method for initializing parameters is a linear combination SM method; Moreover, the split (or merge) of the mean vector is independent of the covariance

matrix, vice versa. Therefore, this is a heuristic procedure without theoretical support. This means that the merge of the mean vector doesn't consider the influence of the covariance matrix, and vice versa. Mean vector and covariance matrix are first order and second order statistical moments, respectively. They are closely related. So it is not suitable for treating them isolatedly. Based on the discussions, we propose new split and merge methods for initializing the parameters of the model.

4.1. Merge Method

From the mixture model (1) and its constrained condition, if the i th and j th component are merged as component i' and other components are unchanged, namely their mixing proportion, mean vector and covariance matrix remain unchanged, we can get that the i' th component has the relationships with the i th and j th component as follows

$$p_{i'} = p_i + p_j \tag{7}$$

$$p_{i'} p_{i'}(\boldsymbol{x}|i') = p_i p_i(\boldsymbol{x}|i) + p_j p_j(\boldsymbol{x}|j) \tag{8}$$

Accordingly, we can obtain that the mean vector and the covariance matrix of the i' th component subject to

$$p_i \boldsymbol{\mu}_{i'} = p_{i'} E_{p_{i'}(\boldsymbol{x}|i')}\{\boldsymbol{x}\} = \int \boldsymbol{x} p_{i'} p_{i'}(\boldsymbol{x}|i') d\boldsymbol{x}$$

$$= p_i \boldsymbol{\mu}_i + p_j \boldsymbol{\mu}_j \tag{9}$$

$$\Sigma_{i'} = E_{p_{i'}(\boldsymbol{x}|i')}\{(\boldsymbol{x}-\boldsymbol{\mu}_{i'})(\boldsymbol{x}-\boldsymbol{\mu}_{i'})^{\mathrm{T}}\} = E_{p_{i'}(\boldsymbol{x}|i')}\{\boldsymbol{x}\boldsymbol{x}^{\mathrm{T}}\} - \boldsymbol{\mu}_{i'}\boldsymbol{\mu}_{i'}^{\mathrm{T}}$$

$$= \frac{p_i}{p_{i'}} E_{p_i(\boldsymbol{x}|i)}\{\boldsymbol{x}\boldsymbol{x}^{\mathrm{T}}\} + \frac{p_j}{p_{i'}} E_{p_j(\boldsymbol{x}|j)}\{\boldsymbol{x}\boldsymbol{x}^{\mathrm{T}}\} - \boldsymbol{\mu}_{i'}\boldsymbol{\mu}_{i'}^{\mathrm{T}}$$

$$= \frac{1}{p_{i'}}\left(p_i \Sigma_i + p_j \Sigma_j + p_i \boldsymbol{\mu}_i (\boldsymbol{\mu}_i)^{\mathrm{T}} + p_j \boldsymbol{\mu}_j (\boldsymbol{\mu}_j)^{\mathrm{T}} - p_{i'} \boldsymbol{\mu}_{i'} \boldsymbol{\mu}_{i'}^{\mathrm{T}}\right) \tag{10}$$

$$p_{i'}(\Sigma_{i'} + \boldsymbol{\mu}_{i'}\boldsymbol{\mu}_{i'}^{\mathrm{T}}) = p_i \Sigma_i + p_j \Sigma_j + p_i \boldsymbol{\mu}_i (\boldsymbol{\mu}_i)^{\mathrm{T}} + p_j \boldsymbol{\mu}_j (\boldsymbol{\mu}_j)^{\mathrm{T}} \tag{11}$$

(7), (9) and (11) give the relationships of the i' th component with the i th and j th components.

4.2. Split methods

The merge method is a well-posed problem. Since the split method is the inverse problem of the merge method, it's ill posed. Based on the fact that the covariance matrix is positive definite symmetric, we consider its orthogonal decomposition, and design some split methods. If the k th component is split as the j' th and k' th component, the corresponding mixing proportions, the mean vectors and the covariance matrices should satisfy the relationships analogous to (7), (9) and (11):

$$p_k = p_{j'} + p_{k'} \tag{12}$$

$$p_k \boldsymbol{\mu}_k = p_j \boldsymbol{\mu}_{j'} + p_k \boldsymbol{\mu}_{k'} \tag{13}$$

$$p_k(\Sigma_k + \boldsymbol{\mu}_k(\boldsymbol{\mu}_k)^{\mathrm{T}}) = p_{j'}(\Sigma_{j'} + p_j \boldsymbol{\mu}_j \boldsymbol{\mu}_{j'}^{\mathrm{T}}) + p_{k'}(\Sigma_{k'} + \boldsymbol{\mu}_k \boldsymbol{\mu}_{k'}^{\mathrm{T}}) \tag{14}$$

The difficulty is that, given the mixing proportion, the mean vector and the covariance matrix of the k th component, when solving the ones of the j' th and the k' th components, there will be infinite sets of solutions because the number of equations is less than the number of the unknowns. It is very difficult to select a set of appropriate solutions. Clearly, the mixing proportion, the mean vector and the covariance matrix obtained in methods by Ueda et al. [14] don't satisfy (12), (13) and (14). Now we discuss how to address this difficulty according to the methods of covariance matrix decomposition.

Split method discussed here is based on the following theorem**:**

Theorem

There exists $A_k = \left(\mathbf{a}_1^{(k)}, \mathbf{a}_2^{(k)}, \cdots \mathbf{a}_D^{(k)}\right)^{\mathrm{T}}$ *s.t.* $\Sigma_k = A_k{}^{\mathrm{T}} A_k$, *and*

$$\boldsymbol{a}_i^{(k)\mathrm{T}} \boldsymbol{a}_j^{(k)} = \begin{cases} 0 & i \neq j \\ l_i & i = j \end{cases}, \tag{15}$$

where $l_j > 0 \ (j = 1,2,\cdots,D)$ *are eigenvalues of the covariance matrix* Σ_k.

According to the theorem above, we discuss split method. Denoting $A_{j'} = [\boldsymbol{a}_1^{(j')}, \boldsymbol{a}_2^{(j')}, \cdots \boldsymbol{a}_D^{(j')}]$, $A_{k'} = [\boldsymbol{a}_1^{(k')}, \boldsymbol{a}_2^{(k')}, \cdots \boldsymbol{a}_D^{(k')}]$, $\Sigma_{j'} = A_{j'} A_{j'}{}^{\mathrm{T}} = \sum_{m=1}^{D} \boldsymbol{a}_m^{(j')} \left(\boldsymbol{a}_m^{(j')}\right)^{\mathrm{T}}$, and $\Sigma_{k'} = A_{k'} A_{k'}{}^{\mathrm{T}} = \sum_{m=1}^{D} \boldsymbol{a}_m^{(k')} \left(\boldsymbol{a}_m^{(k')}\right)^{\mathrm{T}}$, we randomly sample a, b and u_l $(l = 1,2,\cdots D)$ via the Beta distribution

$$a \sim Be(2,2),\ b \sim Be(2,2),$$

$$u_l \sim Be(1,1)\ (l = 1,2,\cdots D)$$

To set

$$p_{j'} = p_k.a\ ,\ p_{k'} = p_k.(1-a) \tag{16}$$

$$\boldsymbol{\mu}_{j'} = \boldsymbol{\mu}_k - \sqrt{\frac{p_{k'}}{p_{j'}}}\sum_{l=1}^{D} u_l \boldsymbol{a}_l^{(k)},\ \boldsymbol{\mu}_{k'} = \boldsymbol{\mu}_k + \sqrt{\frac{p_{j'}}{p_{k'}}}\sum_{l=1}^{D} u_l \boldsymbol{a}_l^{(k)} \tag{17}$$

$$\boldsymbol{a}_l^{(j')} = \sqrt{b(1-u_l^2)\frac{p_k}{p_{j'}}}\boldsymbol{a}_l^{(k)},\ \boldsymbol{a}_l^{(k')} = \sqrt{(1-b)(1-u_l^2)\frac{p_k}{p_{k'}}}\boldsymbol{a}_l^{(k)}.$$
$$(l = 1,2,\cdots D) \tag{18}$$

We can prove the following proposition.

Proposition. *The mixing proportions, the mean vectors and the covariance matrices obtained by the above equations satisfying (12), (13) and (14) and the covariance matrices $\Sigma_{j'}$ and $\Sigma_{k'}$ are positive definite symmetric.*

Now, we give another method for solving $\boldsymbol{\mu}_{j'}$, $\Sigma_{j'}$, $\boldsymbol{\mu}_{k'}$ and $\Sigma_{k'}$. Firstly, randomly sampling l and u from $\{1,2,\cdots,D\}$ and $u \sim Be(1,1)$ respectively, we set

$$\boldsymbol{\mu}_{j'} = \boldsymbol{\mu}_k - \sqrt{\frac{p_{k'}}{p_{j'}}}u\boldsymbol{a}_l^{(k)},\ \boldsymbol{\mu}_{k'} = \boldsymbol{\mu}_k + \sqrt{\frac{p_{j'}}{p_{k'}}}u\boldsymbol{a}_l^{(k)} \tag{19}$$

$$\boldsymbol{a}_m^{(j')} = \begin{cases} \sqrt{b(1-u^2)\dfrac{p_k}{p_{j'}}}\boldsymbol{a}_m^{(k)} & m = l \\ \sqrt{\dfrac{p_{k'}}{p_{j'}}}\boldsymbol{a}_m^{(k)} & m \neq l \end{cases},$$

$$\boldsymbol{a}_m^{(k')} = \begin{cases} \sqrt{(1-b)(1-u^2)\dfrac{p_k}{p_{k'}}}\boldsymbol{a}_m^{(k)} & m = l \\ \sqrt{\dfrac{p_{j'}}{p_{k'}}}\boldsymbol{a}_m^{(k)} & m \neq l \end{cases} \tag{20}$$

Directly by (20), we have

$$\Sigma_{j'} = \sum_{l=1}^{D}\boldsymbol{a}_l^{(j')}\left(\boldsymbol{a}_l^{(j')}\right)^{\mathrm{T}} = \frac{p_{k'}}{p_{j'}}\sum_{\substack{m=1 \\ m\neq l}}^{D}\boldsymbol{a}_m^{(k)}\left(\boldsymbol{a}_m^{(k)}\right)^{\mathrm{T}}$$
$$+ b\left(1-u^2\right)\frac{p_k}{p_{j'}}\boldsymbol{a}_l^{(k)}\left(\boldsymbol{a}_l^{(k)}\right)^{\mathrm{T}}$$
$$= \frac{p_{k'}}{p_{j'}}\Sigma_k + (b - bu^2 - 1)\frac{p_k}{p_{j'}}\boldsymbol{a}_l^{(k)}\left(\boldsymbol{a}_l^{(k)}\right)^{\mathrm{T}} + \boldsymbol{a}_l^{(k)}\left(\boldsymbol{a}_l^{(k)}\right)^{\mathrm{T}} \tag{21}$$

In the same manner, we get

$$\Sigma_{k'} = \frac{p_{j'}}{p_{k'}}\Sigma_k + (bu^2 - b - u^2)\frac{p_k}{p_{k'}}\boldsymbol{a}_l^{(k)}\left(\boldsymbol{a}_l^{(k)}\right)^{\mathrm{T}}$$
$$+ \boldsymbol{a}_l^{(k)}\left(\boldsymbol{a}_l^{(k)}\right)^{\mathrm{T}} \tag{22}$$

Note that the above method for determining $\boldsymbol{\mu}_{j'}$, $\Sigma_{j'}$, $\boldsymbol{\mu}_{k'}$ and $\Sigma_{k'}$ only utilizes the l th column vector $\boldsymbol{a}_l^{(k)}$ of $\boldsymbol{A}_k$. In practical applications, an efficient approach to selecting the candidate column is directly to choose the one corresponding with the maximum eigenvalue of the covariance matrix Σ_k. In next section, we adopt this method. For this method, we can also get the proposition of the previous method.

Note that in one-dimensional data space, the mean vector and covariance matrix degenerate to a real number correspondingly, e.g., $D = 1$, $\Sigma_k = s_k^2$, and $\boldsymbol{A}_k = \boldsymbol{a}^{(k)} = s_k$. The Eqs. (19) and (20) degenerate to

$$\mu_{j'} = \mu_k - \sqrt{\frac{p_{k'}}{p_{j'}}}us_k,\ \mu_{k'} = \mu_k + \sqrt{\frac{p_{j'}}{p_{k'}}}us_k;$$

$$s_{j'}^2 = b(1-u^2)\frac{p_k}{p_{j'}}s_k^2,\ s_{k'}^2 = (1-b)(1-u^2)\frac{p_k}{p_{k'}}s_k^2.$$

Therefore, the algorithms discussed here will be consistent with the split merge method adopted in the RJMCMC algorithm by Richardson and Green [9]. In another words, our methods can extend that of Richardson and Green to high-dimensional data space. This conclusion is excited and very important.

5. Experiments

In this section, we applied the method above described to color image segmentation. In the experiments, we first transform the image from RGB space to YUV space. To speed up computations, we use *k*-means clustering for selecting the initial structure and parameters of the model. For the segmented results obtained by the above method, we make the post-processing via connected component analysis. Figure 1 displays some color image segmentation results, where the final segmentation results are obtained after post-processing.

6. Conclusions

We discuss unsupervised statistical image segmentation based on the Gaussian mixture by combing the VB learning and sequential model selection with the split and merge operations. In future work, we should look for the following aspects:

1. Incorporating he VB method into the data-driven techniques is to study Markov random field for image segmentation.
2. Introducing the split and merge methods proposed in this paper into the RJMCMC methods is to investigate fully Bayesian

analysis of finite mixture model in high-dimensional data space.

Reference

[1] C. Andrieu, P. M. Djuric, & A. Doucet, "Model selection by MCMC computation". *Signal Processing*, 81, 2001.

[2] H. Attial, "A Variational Bayesian framework for graphical models", *Advances in NIPS* 12, 2000.

[3] C. M. Bishop, *Neural Networks for Pattern Recognition*. Oxford University Press, 1995.

[4] E. Clark & A. Quinn, "A data-driven Bayesian sampling scheme for unsupervised image segmentation", *Proc. ICASSP*, 1999.

[5] A. P.Deppster, N. M. Laurd, & D. B. Rubin, "Maximum likelihood from incomplete data via the EM algorithm". *Journal of the Royal Statistical Society Series B*, 1977.

[6] Z. Ghahramani & M. J. Beal, "Variational inference for Bayesian mixture of factor analysers", *Advances in NIPS* 12, 2000.

[7] P. J. Green, "Reversible Jump Markov Chain Monte Carlo Computation and Bayesian Model Determination", *Biometrika*, 82:711-732, 1995.

[8] U. Grenander and M. I. Miller, "Representations of knowledge in complex systems", *Journal of the Royal Statistical Society Series B*, 56, 1994.

[9] S. Richardson, & P. J. Green, "On Bayesian analysis of mixtures with an unknown number of components". *Journal of the Royal Statistical Society Series* B, 59, 1997

[10] S. J. Roberts, D. Husmerier, I. Rezek, & W. Penny, "Bayesian approaches to Gaussian mixture modeling", *IEEE Trans. PAMI*, 20, 1998.

[11] M. Sato, "On-line model selection based on the Variational Bayes", *Neural Computation*, 13(7), 2001.

[12] M. Stephens, "Bayesian analysis of mixtures with an unknown number of components---an alternative to reversible jump methods". *Annals of Statistics*, 28: 40-74, 2000.

[13] Z. Tu, S. C. Zhu, & H. Y. Shum, "Image segmentation by data driven Markov Chain Monte Carlo", *Proc. ICCV*, 2001.

[14] N.Ueda, R. Nakano, Y. Z. Ghahramani, & G. E. Hiton, "SNEM Algorithm for Mixture Models". *Neural Computation*, 12(10), 2000

[15] S. C. Zhu, R. Zhang, & Z. Tu, "Integrating bottom-up for object recognition by data driven Markov Chain Monte Carlo", CVPR, 2000

[16] S. Waterhouse, D. Mackay, & T. Robinson, "Bayesian methods for mixture of experts", *Advances in NIPS* 8, 1996.



Figure 1: Color image segmentation by VBMM. Left: original images; Right: segmentation images.